

Lakshay Chauhan

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EDUCATION

Indraprastha Institute of Information Technology (IIIT) Delhi Jul 2021 – Jun 2025
Bachelor of Technology in Computer Science and Engineering Delhi, India

WORK EXPERIENCE

Member of Technical Staff Jul 2025 – Present
ZL Technologies, Inc / [Website](#) Hyderabad, India

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

Research Assistant | [Gitlab Project Organization](#) May 2024 – May 2025
Networks and Systems Security Lab - IIITD / [Website](#) Delhi, India

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

PROJECTS

AI Customer Service Agent | [Repository](#) May 2025 – Jul 2025

- Built an end-to-end AI-powered customer service agent using Python and LiveKit supporting real-time voice and chat interactions.
- Designed backend workflows for continuous knowledge-base improvement, asynchronous task processing, and conversational context management. Integrated REST/WebSocket APIs and modular service components to enable scalable multi-channel customer support automation.

Real-Estate Match | [Repository](#) Jul 2025 – Sep 2025

- Developed a full-stack real estate platform using Django enabling property discovery and interaction workflows for buyers and sellers.
- Designed relational database schemas and backend APIs for property listings, user management, and matchmaking functionality. Implemented frontend-backend integration, authentication workflows, and scalable CRUD operations for dynamic application features.

Quorum — Distributed Key-Value Store | [Repository](#) Jan 2026 – Mar 2026

- Engineered a distributed key-value store in Java with focus on scalable backend architecture and inter-node communication.
- Implemented data storage and synchronization mechanisms to support distributed systems experimentation and fault-tolerant design concepts.
- Built modular backend components for request handling, persistence, and concurrent access management in a multi-node environment.

TECHNICAL SKILLS

Languages: C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell

Frameworks: PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib

Developer Tools: Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SQLite3

Technical Electives: Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing

AWARDS

- Awarded the **Summer Undergraduate Research Fellowship** (2023) by **IRD-IIITD** for the project “Utilizing Ultrasonic Distance Sensors as a Mapping Tool to Design User-Friendly CST”.
- Awarded the **CHANAKYA Fellowship** (2024) by **iHub Anubhuti Foundation** for the project “A Unified Approach to User Emotion Detection through Emojis and Textual Analysis”.